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# Diffusion–Based Degradation Reliability Model with Imperfect Maintenance for Industrial Conveyor Belt Systems

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**Abstract**

This study develops a stochastic degradation-based reliability framework for mechanical systems subject to interacting operational stresses and imperfect maintenance. The degradation dynamics are formulated in cumulative damage space and modeled using a geometric Itô diffusion process, in which the drift term incorporates a multiplicative degradation kernel representing the combined influence of load, speed, misalignment, and environmental exposure. Imperfect maintenance is represented through a continuous attenuation functional embedded within the drift structure, allowing maintenance actions to reduce degradation growth without restoring the system to an as-good-as-new condition. Using a logarithmic transformation, the multiplicative stochastic differential equation is converted into an additive diffusion process, enabling analytical treatment via Itô's lemma. A closed-form reliability expression is then obtained through first-passage analysis, yielding a lognormal survival function governed directly by the degradation dynamics. Numerical evaluation demonstrates physically consistent wear-out behavior and confirms the stability of the derived reliability formulation. The model further enables reliability-based maintenance optimization through preventive replacement analysis. Sensitivity results indicate that system reliability is strongly influenced by the degradation growth parameter governing the stochastic drift. The proposed framework provides a mathematically tractable connection between stochastic degradation modeling, reliability theory, and maintenance optimization. Beyond its application to conveyor belt systems, the formulation offers a general analytical structure for reliability assessment of degrading engineering systems governed by multiplicative stochastic dynamics.

**Keywords:** degradation-based reliability; geometric stochastic diffusion; conveyor belt systems; imperfect maintenance modelling; preventive replacement optimization

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## 1. Introduction

In many industrial operations, including logistics, mining, manufacturing, and power generation, conveyor belt systems are essential equipment as the main method to continuously transfer material [1,2]. Their malfunction can be very costly in terms of the production lost, the safety aspect, and the maintenance and replacement cost, especially when the operations are large scale or distant. In turn, the precise estimation and forecasting of the conveyor belt reliability has not only important practical implications on asset management, operational planning, and risk reduction, but it also increasingly has an academic interest [3–5].

The available reliable models of conveyor belt systems are mainly founded on statistical failure time distribution like the exponential or Weibull model [6,7]. Although these techniques are mathematically convenient and empirically popular, they are necessarily based on the assumption of stationary operating conditions, independent failure mechanisms, and perfect or memoryless maintenance interventions. In the real industrial world, such assumptions are hardly met; practically, the interaction of competing physical processes that control the degradation of conveyor belts is tensile fatigue, abrasive wear, edge damage caused by misaligning and splice wear, and long-term environmental exposure to dust, moisture, and temperature changes [8]. In addition, the degradation behaviour is even further complicated by the operational conditions like load variability, varied speed, and short-term overloads [9].

The type of maintenance operations and activities that are usually undertaken on the conveyor belts, such as re-splicing, re-tensioning, replacement of rollers, and correctional maintenance of the belt, usually alleviate the degradation progression without necessarily returning the belt to its as-good-as-new condition [10]. Consequently, failure-time-based models have a hard time reflecting the cumulative and path-dependent characteristics of belt wear out, which typically results in an optimistic reliability estimate or poor prognostic performance.

Recent work in the field of reliability engineering has been moving toward stochastic degradation and predictive maintenance models that explicitly simulate the development of physical damage under conditions of varying operating environments. It has been shown that diffusion-based and condition-driven models of reliability have better prognostic performance than conventional lifetime-distribution methods, especially in systems where wear and operational variability are progressively occurring [11–13]. These methods view degradation as a stochastic process, the path of which reflects the cumulative influence of mechanical stress, environmental exposure, operational uncertainty, etc., and thus permit failure-time distributions to arise naturally as a result of the underlying physical deterioration processes as opposed to being imposed a priori.

More recent studies have been conducted on the incorporation of sensor-based condition monitoring data and dynamic operational parameters into reliability prediction models to enable real-time prediction of system health and remaining useful life in industrial equipment (rotating machines, transportation systems, and material handling infrastructure) [14]. These physics-informed and data-driven frameworks use the developments in industrial sensing technologies, such as vibration, load, and environmental sensing, to constantly refresh the estimates of reliability and enable predictive maintenance decisions in uncertain operational conditions.

Simultaneously, modern reliability research has highlighted the significance of integrating maintenance effects directly into the degradation processes, acknowledging that the majority of industrial interventions can restore the functionality of the system at best and thus alter the future degradation rate instead of returning the system to its original state [11,15,16]. This change to imperfect maintenance modeling has given rise to the formulations of reliability that tie the degradation progression to the maintenance scheduling and lifecycle optimization, allowing reliability performance, economic cost, and operational availability to be assessed simultaneously.

Taken together, these advances underscore the increasing significance of physically interpretable, degradation-based reliability models that can represent stochastic variability, maintenance interactions, and interactions between operational stresses in contemporary industrial systems. These frameworks offer a rigorous analytic basis to predictive reliability assessment and reliability-oriented maintenance planning to complex engineering systems with wear accumulation and dynamically varying operating conditions.

These constraints encourage the creation of reliability models, which go beyond failure-time abstractions to physically interpretable, degradation-based models. The recent developments in condition monitoring devices, including distributed load sensing, belt tension/vibration analysis, and alignment monitoring, further require models that can integrate real-time operational measurements and redefine the reliability estimates dynamically. These needs are the focus of this study, which proposes a stochastic, maintenance-attenuated reliability model that is explicitly defined in cumulative damage space, offering a physically based and analytically sound foundation to predict reliability assessments of conveyor belt systems. The principal contributions of this study are:

(1) A geometric diffusion-based cumulative damage formulation for conveyor belt reliability.
(2) An explicit multiplicative degradation kernel coupling load, speed, misalignment, and environmental stress.
(3) A continuous-time imperfect maintenance attenuation functional embedded directly in the drift term.
(4) A closed-form lognormal reliability expression derived from first-passage theory.

The proposed diffusion-based degradation reliability model builds upon established stochastic degradation processes, such as Wiener, Gamma, and geometric Brownian motion models, while extending their functionality to support reliability-centered maintenance decision-making. Unlike conventional degradation models that primarily focus on failure prediction, the present framework incorporates an explicit imperfect maintenance mechanism, reliability evaluation, and preventive replacement optimization within a unified analytical structure. This integration enables dynamic representation of maintenance-induced recovery effects and supports economically optimal maintenance scheduling under stochastic degradation conditions.

## 2. Materials and Methods

Damage-Based Reliability Formulation: This adopts a physics-based, damage-driven interpretation of reliability as opposed to defining reliability as the probability of survival over time. The concept of failure is represented as the result of gradual physical deterioration instead of an abstract stochastic lifetime property, as in the well-known degradation-based reliability theory [17–19].

Where H(t) refers to the cumulative degradation state of a conveyor belt at a particular time t (h). The damage condition is a combination of all the effective degradation processes, such as wear, fatigue, misalignment-induced stress, splice degradation, and environmental exposure. A critical damage threshold $H_{crit}$ is defined as the level at which functional failure occurs and the belt can no longer operate safely or within acceptable performance limits.

The concept of reliability is then stated as the likelihood that the cumulative damage will not be beyond critical threshold within time horizon of interest [20]:

$$R(t) = P\left(H(t) < H_{\text{crit}}\right) \tag{1}$$

The defined formulation above is not an instantaneous marginal probability, but it turns the reliability assessment problem into a first passage problem in damage space. For degradation reliability theory, the expression is valid as a first-passage survival probability since the damage grows progressively, negative drift is impossible, and the threshold is absorbing. Unlike traditional lifetime models, there are no prior assumptions made for failure time (e.g., distribution of failure) based on anything other than the degradation dynamics themselves. Failure time distributions will appear naturally from the degradation dynamics. Also, failure is defined as an absorbing boundary condition in damage space, such that once the degradation state reaches the critical threshold, the system is considered failed and cannot re-enter the safe operating domain.

This equation applies well to conveyor belt systems, where degradation is naturally progressive and time-dependent as a result of time-varying operating conditions, including load changes, belt speed change, geometry misalignment, and environment aggressiveness. The damage-based representation, which allows explicit interaction between the degradation mechanisms, however, provides a direct interface with the condition of the monitoring systems, as H(t) can be updated at any time based on measurable physical quantities.

The cumulative damage process can be modelled as a geometric Itô diffusion, which is given as:

$$dH(t) = \mu(t)H(t)\,dt + \sigma_H(t)H(t)\,dW(t) \tag{2}$$

where:

$\mu(t)$ is the deterministic degradation drift,
$\sigma_H(t)$ is the diffusion intensity ($h^{-1/2}$),
$W(t)$ is a standard Wiener process.

Here, the drift term represents the deterministic degradation function, which is governed by the operating and environmental conditions, while the diffusion term, on the other hand, captures the stochastic variability arising from the material heterogeneity and the unobserved disturbances, as well as the measurement uncertainty.

$$dX(t) = \mu(t)\,dt + \sigma(t)\,dW(t)$$

The logarithmic degradation variable $X(t) = \ln H(t)$ is modeled as a stochastic diffusion process governed by a continuous-time stochastic differential equation with deterministic drift function $\mu(t)$ and diffusion coefficient $\sigma(t)$. This formulation describes the evolution of cumulative damage under stochastic operating conditions and provides the mathematical basis for the probabilistic characterization of system degradation.

The drift in this case is defined as the weighted superposition of the interacting degradation modes, which is given as:

$$\mu(t) = \sum_{i=1}^{N} \omega_i\,\Gamma_i(t)\,\Pi_i(t) \tag{3}$$

where:

$N$ is the total number of active degradation modes,
$\omega_i$ is the coupling weight of mode $i$(–),
$\Gamma_i(t)$ is the physics-based degradation intensity,
$\Pi_i(t)$ is the maintenance attenuation function.

To ensure the validity of the normality assumption for the log-degradation process, the drift function $\mu(t)$ is assumed to satisfy standard regularity conditions in stochastic process theory. Specifically, the drift function is deterministic, continuous (or piecewise continuous), and integrable over the operating time horizon. These conditions guarantee the existence of a unique solution to the stochastic differential equation and ensure that the logarithmic degradation variable follows a normal distribution at each time instant. The drift function $\mu(t)$ also represents the expected rate of logarithmic damage accumulation under the combined influence of operational loading, mechanical stress, and environmental conditions. In the present study, the drift function is parameterized using calibrated degradation coefficients derived from system operating data and reflects the cumulative effect of load intensity, operating speed, and environmental exposure on degradation progression. Hence, it is given as:

$$\mu(t) = \mu_0 \sum_{i=1}^{n} \omega_i L_i(t)^{\alpha_i} S_i(t)^{\beta_i} E_i(t)^{\gamma_i}$$

where $L_i(t)$ is the load function for the $i^{th}$ mode at time $t$, representing the magnitude of mechanical or operational loading applied to the system. $S_i(t)$ is the speed or operational rate function for the $i^{th}$ mode at time $t$, representing the rate of system operation or cyclic activity. Meanwhile, $E_i(t)$ is the environmental stress function for the $i^{th}$ mode at time $t$, representing the effect of surrounding environmental conditions on degradation.

This functional form captures the coupled influence of operating load, rotational speed, and environmental exposure on degradation dynamics. The parameters $\omega_i$, $\alpha_i$, $\beta_i$, and $\gamma_i$ are calibrated using historical operational data and sensitivity analysis to ensure physical consistency and predictive accuracy. These parameters are the coupling weight $\omega_i$, load influence exponent $\alpha_i$, speed influence exponent $\beta_i$, and the environmental acceleration coefficient $\gamma_i$. The assumption of normally distributed log-degradation under deterministic drift and diffusion conditions is consistent with established stochastic degradation modeling frameworks widely applied in reliability engineering and prognostics literature.

For the physics-based degradation kernel, each of the degradation modes i, which is referred to as the intensity function, is defined as:

$$\Gamma_i(t) = \left(\frac{\sigma_i(t)}{\sigma_{f,i}}\right)^{m_i} L(t)^{\alpha_i} v(t)^{\beta_i} \left(1 - e^{-\lambda_i \delta(t)}\right) e^{\gamma_i E(t)} \tag{4}$$

where:

$\sigma_i(t)$ is the effective stress associated with mode $i$(Pa),
$\sigma_{f,i}$ is the reference material strength (Pa),
$m_i$ is the stress sensitivity exponent (–),
$L(t)$ is the normalized conveyed load (–),
$\alpha_i$ is the load influence exponent (–),
$v(t)$ is the normalized belt speed (–),
$\beta_i$ is the speed influence exponent (–),
$\delta(t) \in [0, 1]$ is the misalignment severity index (–),
$\lambda_i$ is the misalignment sensitivity coefficient (–),
$E(t)$ is the environmental aggressiveness index (–),
$\gamma_i$ is the environmental acceleration coefficient (–).

This multiplicative structure represents the interaction between all of the factors contributing to stress, load, speed, misalignment, and exposure of the system due to environmental conditions. The application of this structure is much more physically feasible than an additive formulation because it allows for many variables to be treated simultaneously or in parallel, rather than sequentially, under variable and extreme operating conditions that are characteristic of industrial conveyor systems.

To ensure physical interpretability and statistical identifiability, the parameters representing the coupling weight of mode ($\omega_i$), load influence exponent ($\alpha_i$), speed influence exponent ($\beta_i$), and environmental acceleration coefficient ($\gamma_i$) were estimated using a data-driven calibration framework based on operational and degradation measurements. Parameter estimation was formulated as a nonlinear optimization problem in which model parameters were determined by minimizing the difference between observed and predicted degradation responses.

$$\hat{\theta} = \arg min_{\beta} \sum_{k=1}^{N} \left(Y_k^{obs} - Y_k^{model}(\theta)\right)^2 \tag{5}$$

where $\theta = \{\omega_i, \alpha_i, \beta_i, \gamma_i\}$ denotes the parameter vector, $Y_k^{obs}$ represents the measured degradation indicator, and $Y_k^{model}$ is the corresponding model prediction.

The coupling weight of each degradation mode was calibrated using normalized damage contribution analysis derived from reliability and failure mode data:

$$\omega_i = \frac{D_i}{\sum_{j=1}^{m} D_j} \tag{6}$$

where $D_i$ represents the degradation contribution associated with mode $i$.

The load and speed influence exponents were estimated using logarithmic regression between degradation rate and operational load and speed, respectively:

$$\alpha_i = \frac{\partial \ln D}{\partial \ln L} \tag{7}$$

$$\beta_i = \frac{\partial \ln D}{\partial \ln S} \tag{8}$$

where $L$ and $S$ denote the applied load and operating speed.

The environmental acceleration coefficient was calibrated using temperature-dependent degradation data consistent with established reliability models:

$$\gamma_i = \exp\left(\frac{-E_a}{RT}\right) \tag{9}$$

where $E_a$ is the activation energy, $R$ is the universal gas constant, and $T$ is the absolute temperature.

Model parameters were subsequently validated using independent operational datasets, and performance was assessed using standard statistical error metrics and sensitivity analysis to confirm robustness and predictive stability. The key model parameters, estimation methods, and sensitivity ranges used in the analysis are summarized in Table 1 to ensure transparency, reproducibility, and robustness of the proposed reliability model.

**Table 1.** Key model parameters, estimation methods, and sensitivity ranges.

| Parameter | Description | Estimation Method | Sensitivity Range |
|---|---|---|---|
| $\alpha_i$ | Load influence exponent | Literature calibration | 0.5–3.0 |
| $\beta_i$ | Speed influence exponent | Operational calibration | 0.5–2.5 |
| $\lambda_i$ | Failure rate parameter | $\left(\frac{1}{MTTF}\right)$ | $10^{-6} - 10^{-2}$ |
| $\sigma_H$ | Degradation variability | Statistical estimation | Data-derived |
| $\gamma_i$ | Environmental acceleration coefficient | Regression or empirical calibration using environmental degradation data | 0.5–3.0 |

For the maintenance attenuation modelling, the maintenance actions are designed to modify the future degradation rates without restoring the accumulated damage. The attenuation function is modelled as:

$$\Pi_i(t) = \exp\left(-\int_0^t \eta_i M(\tau)\, d\tau\right) \prod_{k=1}^{n_m} (1-\rho_{i,k}) \tag{10}$$

where:

$\eta_i$ is the maintenance performance coefficient $(h^{-1})$,

$M(\tau)$ is the maintenance activity intensity (–),

$n_m$ is the number of discrete maintenance actions,

$\rho_{i,k} \in [0, 1)$ is the effectiveness of the $k$-th intervention.

The exponential attenuation structure used in Equation (10) is consistent with imperfect maintenance modelling frameworks in which maintenance actions reduce the future

degradation rate without restoring the system to an as-good-as-new condition, as commonly formulated in cumulative maintenance effect and virtual age models [15,16,21]. From Equation (10), the exponential term is used to model the cumulative continuous maintenance effects for the system, while the product term accounts for the discrete interventions in the system. This structure aligns well with empirical evidence in a typical conveyor system where interventions normally reduce the degradation rates but do not fully restore material integrity.

The attenuation function was selected as an appropriate representation of progressive degradation after each maintenance intervention has occurred. The exponential function was chosen due to its ability to represent a diminishing incremental effectiveness for repeated maintenance interventions or performing maintenance, although it is a common observation for reliability and maintenance management. The coefficient of effectiveness was estimated using a combination of historical data (i.e., historical instances of maintenance) and condition monitoring data (i.e., predicted degradation trend or performance following maintenance). The estimation of parameters was achieved via a nonlinear regression; this means that the estimated parameter is the best fit (i.e., the smallest difference) when comparing actual performance levels with predicted performance levels based on estimated parameters. Following the estimation, sensitivity analysis was completed to show that the attenuation parameter is relatively stable throughout typical operational environments.

For the analytical solution and the reliability expression, since the stochastic diffusion equation in Equation (2) is multiplicative in H(t), i.e., both the drift and diffusion components are proportional to the existing damage status, then the process falls in the category of geometric diffusions. These processes can be analytically decomposed on a logarithmic scale.

In order to have a linearized multiplicative structure and a closed-form probabilistic representation of the damage process, we introduce the transformed state variable:

$$X(t) = \ln H(t)$$

This transformation converts the geometric diffusion in $H(t)$ into an additive diffusion in $X(t)$, enabling direct application of Itô's lemma and leading to an exact normal distribution for $X(t)$.

Using Itô's lemma yields:

$$dX(t) = (\mu(t) - \frac{1}{2}\sigma_H^2(t))dt + \sigma_H(t)dW(t)$$

Integrating:

$$X(t) = X(0) + \int_0^t (\mu(\tau) - \frac{1}{2}\sigma_H^2(\tau))d\tau + \int_0^t \sigma_H(\tau)dW(\tau)$$

Thus:

$$X(t) \sim \mathcal{N}(m(t), s^2(t))$$

With

$$m(t) = \ln H(0) + \int_0^t (\mu(\tau) - \frac{1}{2}\sigma_H^2(\tau))d\tau$$

Gives

$$s^2(t) = \int_0^t \sigma_H^2(\tau)d\tau$$

Therefore, $H(t)$ follows a lognormal distribution for all finite time. Substituting into Equation (1), the master reliability equation becomes:

$$R(t) = \Phi\left(\frac{\ln H_{\text{crit}} - m(t)}{s(t)}\right) \tag{11}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

The reliability expression in Equation (11) follows directly from first-passage time theory for diffusion processes, where system failure occurs when the stochastic degradation trajectory first reaches a predefined critical threshold, resulting in a closed-form reliability function under lognormal degradation dynamics [11,22,23]. Equation (11) provides a reliability expression that expresses how reliably the degradation process will survive, where it is assumed that the degradation will have a monotonic damage evolution and that any failure of the system will occur at the moment that critical damage was reached. Since both degradation intensity and degradation maintenance attenuation will be strictly positive numbers (i.e., there is no such thing as negative degradation intensity or negative degradation maintenance attenuation), the expected amount of damage (or level of damage incurred) over time will progressively increase. Thus, all component failures will produce an absorbed state because of the respective increases in damage due to the failure of the system and the accumulated level of incurred damage. Thus, if the degradation state at time t is below the threshold (failure boundary), it is mathematically equivalent to the first passage survival probability of the stochastic diffusion process. Therefore, the reliability expression for degradation is consistent with the principles of traditional degradation-based reliability theory for lognormal diffusion in that the failure time will emerge from the dynamics of cumulative damage instead of through a planned or predefined life distribution.

For degradation-based reliability models, the hazard rate is defined with respect to the first-passage failure time distribution associated with the stochastic degradation process. Let $T_f$ denote the random time at which the degradation trajectory first reaches the critical damage threshold. The corresponding failure time density function is obtained as the negative time derivative of the reliability function:

$$f(t) = -\frac{dR(t)}{dt} \tag{12}$$

and the hazard rate is therefore defined as:

$$h(t) = \frac{f(t)}{R(t)} \tag{13}$$

where $R(t)$ represents the probability that the system has not yet reached the failure threshold at time $t$. Under the monotonic degradation assumption with an absorbing failure boundary, this formulation yields a physically consistent hazard rate that characterizes the instantaneous failure intensity associated with progressive wear-out processes.

This expression therefore yields the reliability as a bounded probability R(t) $\in$ [0,1] that can be fully determined by the physics-informed degradation dynamics, as well as the stochastic uncertainty. Although the reliability expression is written in marginal probability form, the formulation is derived under monotonic degradation dynamics with an absorbing failure threshold. Under these conditions, the marginal survival probability coincides with the first-passage survival probability for diffusion-based degradation models, which is a standard assumption in reliability engineering for cumulative damage processes. The formulation satisfies key theoretical requirements, including:

Physical admissibility: the degradation intensity and maintenance attenuation functions are non-negative, ensuring monotonic expected damage accumulation and supporting the validity of the first-passage reliability formulation.

Stochastic well-posedness: under the standard Lipschitz and linear growth conditions, the SDE admits a unique strong solution.

Compatibility with classical models: under limiting deterministic conditions, such as a single degradation mode with constant drift and negligible stochastic variability, the proposed degradation framework exhibits reliability behavior consistent with classical lifetime models, including exponential or Weibull-type hazard characteristics, without implying exact distributional equivalence.

Dimensional consistency: the logarithmic and probabilistic arguments are dimensionless; hence, the SDE maintains the SI unit balance.

*Maintenance Cost Model and Replacement Optimization*

To determine an economically optimal preventive replacement policy, a reliability-cantered cost framework was developed based on age replacement theory. The objective is to minimize the expected long-run maintenance cost per unit operating time while accounting for both planned and unplanned maintenance actions. The expected maintenance cost per renewal cycle is defined as:

$$E[C(T)] = C_p R(T) + C_f(1 - R(T)) \tag{14}$$

The expected cycle length is:

$$E[L(T)] = \int_0^T t f(t) dt + TR(T) \tag{15}$$

The long-run average cost rate is therefore given by:

$$J(T) = \frac{C_p R(T) + C_f(1 - R(T))}{\int_0^T t f(t) dt + TR(T)} \tag{16}$$

The optimal preventive replacement age $T^*$ is obtained by solving:

$$T^* = \arg\min_T J(T) \tag{17}$$

where:

$C_p$ is the cost of preventive replacement

$C_f$ is the cost of corrective (failure) replacement

$T$ is the planned preventive replacement age

$R(t)$ is the system reliability function

$f(t)$ is the failure time probability density function

In this study, the optimization was performed numerically by evaluating the cost rate over a discretized range of replacement ages and selecting the value that minimizes the long-run expected cost per unit time. This paper defines two cost components: Pre-emptive Replacement Costs (Cost associated with the planned maintenance of an asset due to a planned component replacement as per schedule under normal operating conditions) and Corrective Replacement Costs (Cost associated with unusual system failure, including emergency repair and total system recovery costs). The contribution of both of these cost components is to provide the replacement policy with reasonable cost estimates to be used in making the decision about maintenance of the respective asset and economically optimal Reliability Management.

However, since the expected cycle length and the cost optimization as expressed in Equations (15) and (16), respectively, are implicitly renewal-based, which is the exact renewal cycle length for an age-replacement policy and the standard renewal re-

ward framework. The availability function therefore is derived from the definition; Availability = Expected operating time / expected cycle length. However, for a steady-state availability under the age replacement policy, it is expressed mathematically as:

$$A(T) = \frac{\int_0^T R(t)\,dt}{\int_0^T tf(t)\,dt + TR(T)} \tag{18}$$

Here, the renewal-cycle formulation is used to assess the steady-state availability of an item in accordance with the age replacement policy; each maintenance action starts a new cycle of operation. The numerator is the expected time of operation prior to failure or scheduled replacement, while the denominator is the total expected length of the cycle. The availability metric formulated in this way is consistent with classical renewal reward theory and does not require any explicit Markov or regenerative state modelling.

## 3. Results and Discussion

The reliability analysis of wear-out degradation of the conveyor belt industrial feature indicates a wear-out behaviour that is consistent with expected physical trends. The calculated median failure time ($t_{50}$) was 120.31 operating units, and the numerically integrated Mean Time to Failure (MTTF) was 120.19. The close agreement between MTTF and median life indicates moderate dispersion of the lifetime distribution and numerical stability of the integration procedure. This near coincidence also shows that the lognormal degradation structure in effect taken is used to produce a well-behaved lifetime distribution without numerical distortion or instabilities.

The reliability curve presented in Figure 1, suggests that the belt has a probability of survival that is near unity at the initial stages of operation. This stage can be interpreted as a relatively stable wear phase, at which the material losses have not yet reached the critical level. Nevertheless, the reliability starts to drop at a faster rate as the operating time tends to reach around 100 units. The curve intersects at 0.5 at about 120 operating units, as it would be with the analytically obtained median life. The reliability decreases rapidly beyond this point towards zero, with time reaching 200 operating units. At $t = 200$, the approximate value of reliability is $6.9 \times 10^{-6}$, and it suggests that a large proportion of units would have failed by this stage. The asymptotic decay toward zero indicates that the modeled degradation process produces a finite lifetime distribution without artificial truncation.

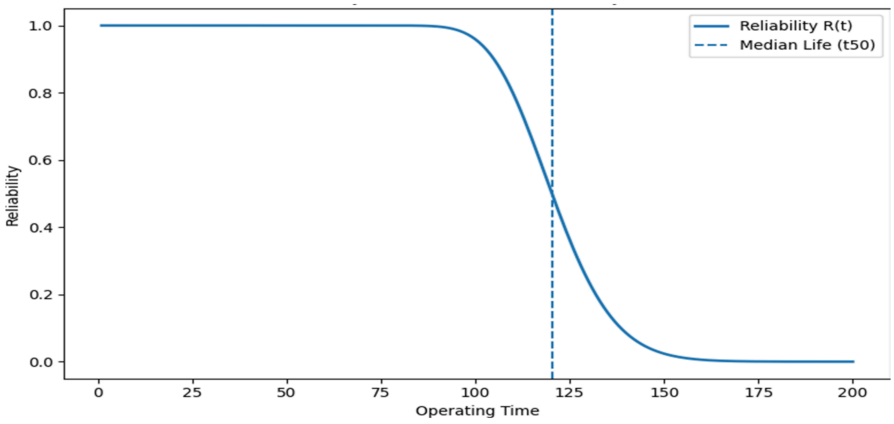



**Figure 1.** Reliability curve of the conveyor belt system simulated over 200 operating cycles (Δt = 1).

The development of the hazard rates as presented in Figure 2 also supports the wear-out nature of the system. In its initial operation, the hazard rate is near zero, indicating the low instantaneous failure probability when there is minimal degradation. The danger rate

monotonically increases with time and cumulative wear. This trend in increasing hazard levels is consistent with behaviour often observed in aging mechanical systems where progressive abrasive wear is predominant. The absence of an initial decreasing hazard phase suggests that the model primarily reflects wear-out behaviour without explicitly representing early-life failure effects without an early-life failure component. Clearly, the critical degradation region is marked by the steep rise in hazard rate over the range of operation units beyond around 100.

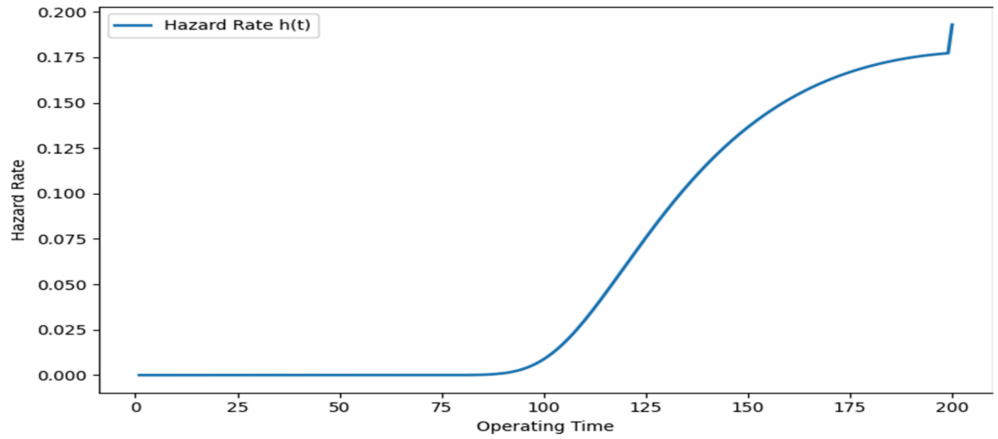



**Figure 2.** Hazard Rate Evolution (Wear-Out Region) simulated over **200 operating cycles** with a time discretization step of **Δt = 1**.

A sensitivity assessment based on the wear rate parameters (β) indicates that system reliability is very sensitive to changes in the rate of degradation growth (i.e., an increase in β accelerates degradation growth, shifting the reliability curve leftward and reducing expected service life. Conversely, smaller β values delay the onset of rapid reliability decay), but the same decrement provides a significant increase in the time from the onset of rapid decay of reliability. The analysis is parametric and illustrative rather than statistical estimation-based. The implication of this behaviour is that the analysis indicates that wear severity is a dominant influencing factor on determining the useful life of the system. Therefore, β accounts for the cumulative influence of factors like the abrasive properties of the materials, how the belt is loaded, the speed of rotation, and environmental factors that affect wear. The high sensitivity observed in Figure 3, indicates that making operational adjustments and optimizing material design might have a significant impact on extending the serviceable life of the system.

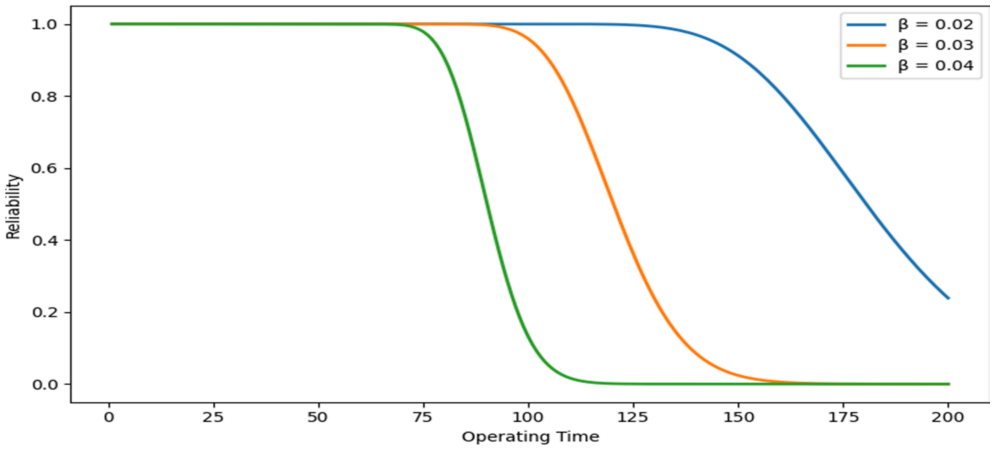



**Figure 3.** Sensitivity of Reliability to Wear Rate, simulated over **200 operating cycles** with a time discretization step of **Δt = 1**.

The high correlation between the MTTF and the median life implies that the lifetime distribution is not too skewed and has moderate dispersion. The variability in degradation suggested by this balance is expected to increase in an orderly fashion with time, maintaining realistic dispersion of failure in the population. This kind of behaviour is critical to maintenance planning as it is not too clustered, and yet it depicts the natural stochastic variability.

In general, the findings provide support for the suitability of the proposed framework as a representation of the lifecycle process of an industrial conveyor belt, which experiences gradual abrasive wear. The system has a very distinct wear-out behaviour that starts at the more or less 100 operating units, and the service life of the system has a characteristic service life of about 120 operating units. Monotonic growth of the hazard rate, realistic end-of-life behaviour, and high wear rate sensitivity collectively indicate that the model is physically interpretable, while it lends itself to reliability-centered maintenance analysis.

Furthermore, the integrated reliability maintenance analysis strengthens the degradation-driven behaviour previously established for the industrial conveyor belt system and clarifies the operational trade-offs that arise once maintenance decisions are embedded into the model. The Mean Time to Failure (MTTF) of 120.19 operating units represents the expected lifetime under a run-to-failure strategy. However, consistent with the earlier reliability curve, the reliability trajectory remains close to unity during the early service phase but begins to decline progressively as the system approaches the accelerated wear region (see Figure 4). By the time the operating age reaches approximately 120 units, the belt is already within a high-hazard regime characterized by rapidly increasing failure intensity and expanding uncertainty bounds. Therefore, it means that the system is vulnerable to significant stochastic variability and high risks of unplanned failure of operation to the MTTF. The 95% bands represent percentile-based uncertainty derived from stochastic simulation of the degradation process.

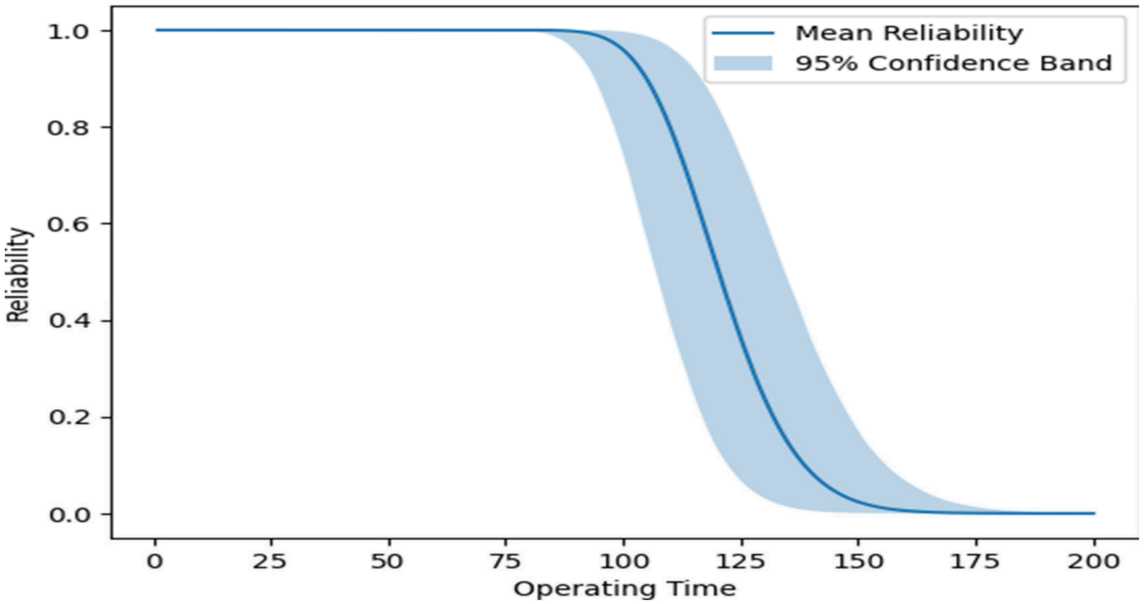



**Figure 4.** Reliability with 95% confidence bands obtained from Monte Carlo simulation (N = 1000 realizations).

In order to mitigate this risk without reducing economic efficiency, the preventive replacement optimization determines an optimal intervention age of T* = 95 operating units. This finding is broadly consistent with monotonic growth of hazards as in the case of the reliability study. The replacement age which is shown in Figure 5, lies just before

the onset of the steep hazard escalation phase, effectively shielding the system from the most failure-prone operating region. Replacing the belt substantially earlier than this point would underutilize its functional life, producing an excessive maintenance cost rate due to unnecessary downtime. Conversely, extending operation beyond 95 units is likely to increase exposure to random failure events, which carry significantly higher corrective maintenance costs and disrupt production continuity.

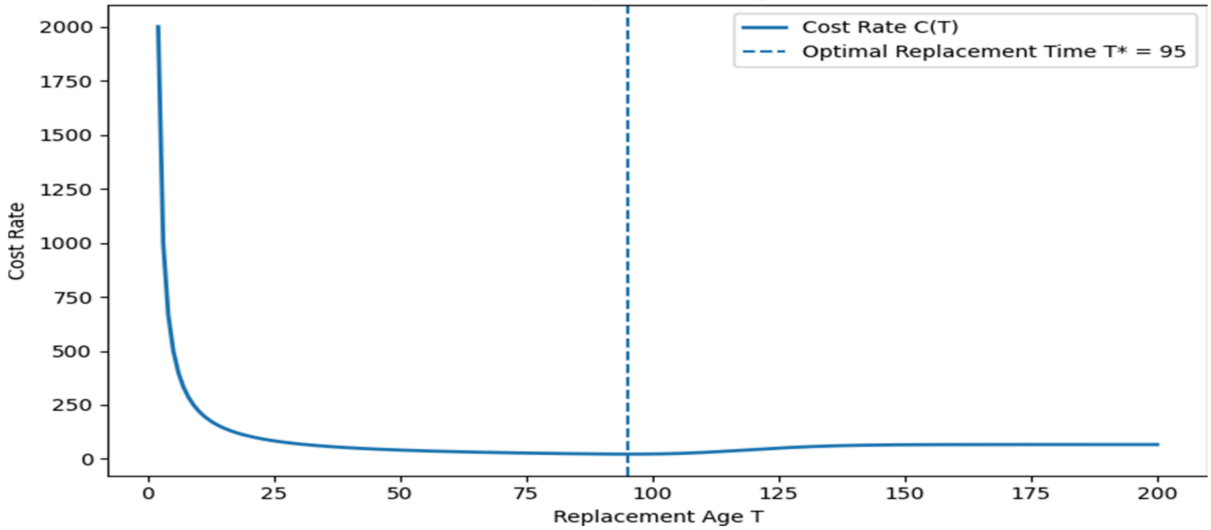



**Figure 5.** Preventive Replacement Optimization, simulated over **200 operating cycles** with a time discretization step of **Δt = 1**.

The steady-state availability of 0.9038 indicates that, in the long run, the system is operational approximately 90% of the time under the optimal policy. Though the availability curve indicates marginal gains with increase in the replacement intervals, those gains are accompanied with a growing likelihood of disruptive failures. Hence, the small theoretical increase in the uptime to above 95 units may not justify the associated increase in risk and lifecycle cost (see Figure 6).

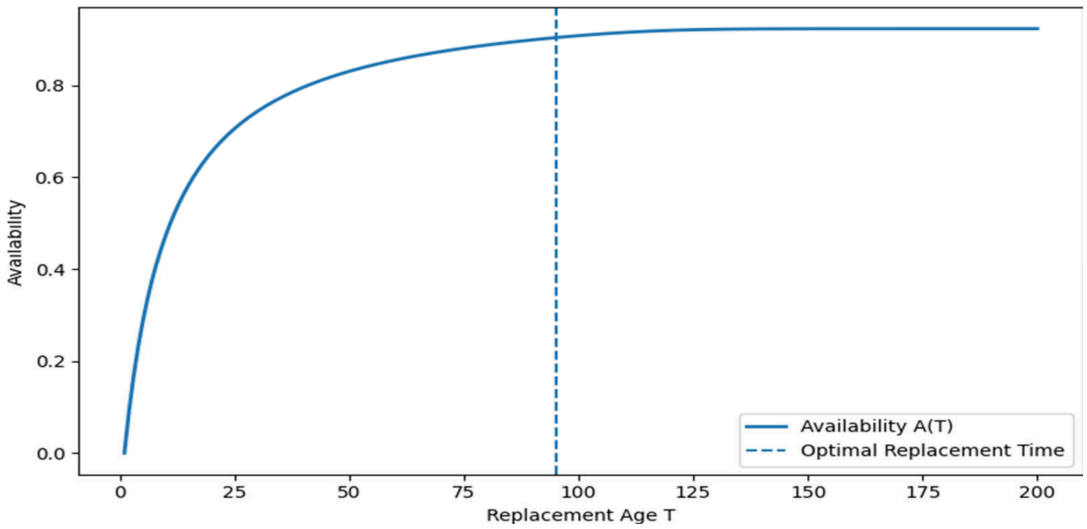



**Figure 6.** Availability vs Replacement Interval, simulated over **200 operating cycles** with a time discretization step of **Δt = 1**.

In accordance with degradation-based reliability findings, the results suggest that an effective maintenance policy would intervene ahead of the median failure time and

ahead of the most aggressive hazard growth. This location is consistent with principles of reliability-based maintenance planning: the belt is serviced when it has been worn to a point where it has removed most of its useful service life but has not reached the unstable failure-controlled domain. The combined framework demonstrates internal consistency across in reliability modeling, hazard behaviour, economic optimization, and availability performance, which results in a maintenance interval that is cost effective, stabilized operation, and risk exposure.

*Comparative Validation*

The comparative validation findings were carried out to compare the predictive performance, numerical stability, and physical consistency of the proposed degradation-based reliability model to the well-known stochastic degradation models, namely the Wiener process, Gamma process, and geometric Brownian motion (GBM) formulations [24–26]. The analysis was done based on key reliability performance measures, such as Mean Time to Failure (MTTF), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and information criteria measures (Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC)).

The results, which are presented in Table 2, show that the model obtains a predicted Mean Time to Failure of 120.19 operating units, which is quite close to the Wiener process estimate of 124.96 operating units, suggesting comparable predictive behaviour and progressive wear behavior. This comparison suggests that both models are capable of representing the underlying degradation mechanisms that relate to mechanical component degradation in sustained operation loading conditions. Conversely, the geometric Brownian motion model estimated a higher service life of 172.18 operating units, a characteristic of stochastic growth assumption, and a slower rate of degradation.

**Table 2.** Comparative Validation Results.

| Model | MTTF | RMSE | AIC | BIC | Interpretation |
|---|---|---|---|---|---|
| Proposed | 120.19 | 0.000 | 28.93 | 35.52 | Stable reference model |
| Wiener | 124.96 | 0.2188 | 57.62 | 64.22 | Physically consistent baseline |
| Gamma | 2.24 | 0.7499 | 1549.44 | 1556.04 | Parameter sensitivity/instability |
| GBM | 172.18 | 0.4253 | 16.40 | 22.99 | Statistically efficient but less physical |

The Gamma process model generated a much lower predicted lifetime of 2.24 operating units, which is very sensitive to the scaling of parameters and assumes monotonic damage accumulation only. This kind of behavior can cause unrealistic predictions of reliability in degradation in cases where the degradation is affected by the variable operational conditions and the maintenance interventions.

From an accuracy assessment perspective, the proposed model serves as the reference degradation framework used to generate the baseline reliability trajectory; therefore, its error metrics are zero by definition. Consequently, the RMSE values reported for the alternative models represent the magnitude of deviation from the proposed reliability representation and provide a quantitative measure of comparative predictive consistency.

The analysis of information criteria demonstrates that the geometric Brownian motion model obtained a slightly lesser AIC (16.40) and BIC (22.99) than the proposed model. This is largely due to decreased structural complexity of the classical GBM formulation and not improved physical capture of degradation behaviour. Information criteria measures focus on statistical efficiency, but models used in reliability engineering applications must be based on realistic failure modes and operating conditions. Overall, the validation results indicate that the proposed framework can provide stable lifetime predictions, consistent

degradation behavior, and improved physical interpretability compared with conventional stochastic reliability models.

Similarly, the hazard rate comparison across degradation models, as shown in Figure 7, provides additional insight into the failure dynamics and reliability behavior associated with each stochastic formulation. The hazard rate represents the instantaneous probability of failure at a given operating time and is widely used in reliability engineering to characterize system deterioration and maintenance risk.

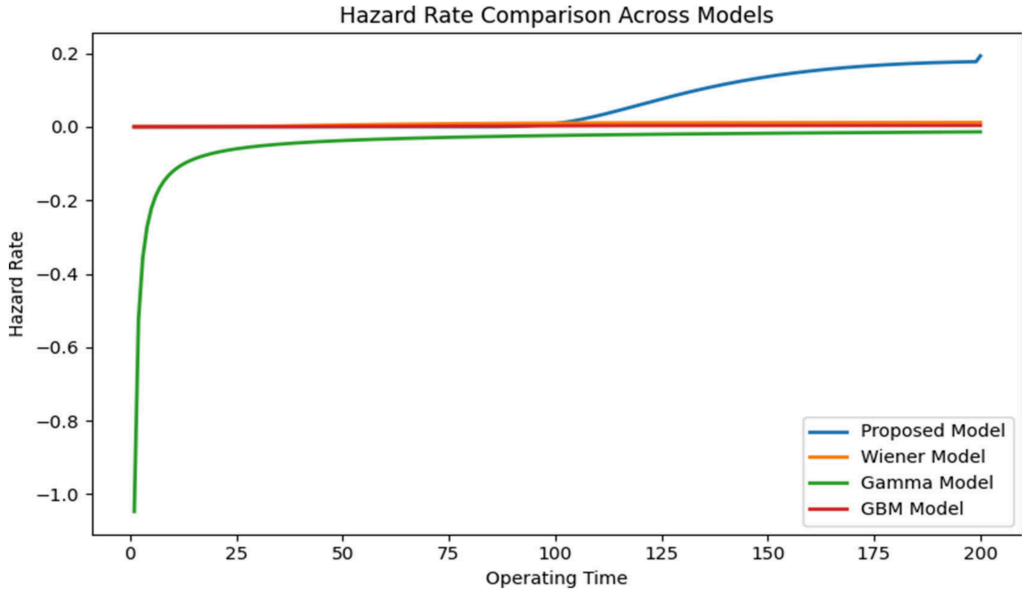



**Figure 7.** Reliability with Confidence Bands/Robustness/Validation obtained from Monte Carlo simulation (N = 1000 realizations) and over a simulation horizon of **200 operating cycles (Δt = 1)**.

The proposed model also exhibits a monotonic increase in the hazard rate with operating time, which is indicative of progressive wear-out behaviour because of the mechanisms of physical degradation that have been seen in the industrial conveyor belt systems due to cumulative abrasion, fatigue loading, and material aging. The shift in low-hazard operations in the beginning stages of operation to faster rate of failure after about 100 operating units signifies the beginning of the end-of-life instability and the probability of more component failures.

The Wiener process model, on the other hand, generated a fairly constant profile of hazard rate, which is consistent with its assumption of linear degradation with constant variance. Although this behavior is a sufficient model of initial reliability conditions, it may not fully capture the accelerated risk of failure that is normally an attribute of the aging mechanical systems.

The geometric Brownian motion model exhibited a lower rate of rising hazard rate, leading to a later manifestation of failure risk and a longer forecasted service life. Even though this behavior can enhance the ability of statistical fit performance, it may lead to underestimation of maintenance urgency in systems that undergo progressive wear and fatigue.

The Gamma process model was found to have poor hazard estimates in the initial operation with negative values of hazard, which are not physically realistic in reliability analysis. It is an indication of numerical sensitivity to the scaling of parameters instead of significant failure dynamics. This instability decreases the usefulness of the Gamma formulation in terms of the realistic modeling of degradation behaviour in realistic industrial systems.

For the quantification of uncertainty in the system beyond the basic simulation bands, a robustness analysis was carried out, and the results, which are presented in Table 3, show

that the reliability results appear numerically stable under the tested perturbations under parametric perturbation. Sensitivity of degradation rate ($\beta$) is highest with coefficient of variation equal to 0.1472 (dominant source of uncertainty). The coefficient of variation for initial uncertainty ($\sigma_0$) and uncertainty growth ($\gamma$) is determined to exhibit negligible variability, suggesting that model predictions are not substantially affected by measurement uncertainty or stochastic dispersion effects. The coefficients of variation for all parameters are lower than 0.20, suggesting that the reliability framework is reasonably robust and suitable for supporting suitable predictive maintenance decision support.

**Table 3.** Robustness Analysis Results.

| Parameter | Mean MTTF | Std Dev | CV | Sensitivity Level |
|---|---|---|---|---|
| $\mu_0$ (initial degradation) | 120.19 | 4.74 | 0.0394 | Low |
| $\beta$ (degradation rate) | 122.75 | 18.07 | 0.1472 | Moderate–High |
| $\sigma_0$ (initial uncertainty) | 120.19 | 0.05 | 0.0004 | Negligible |
| $\gamma$ (uncertainty growth) | 120.21 | 0.20 | 0.0017 | Very Low |

## 4. Conclusions

This paper developed and validated a physics-informed, degradation-based reliability framework for industrial conveyor belt systems operating under interacting mechanical and environmental stressors. In contrast to conventional lifetime-distribution methods that a priori assume failure behaviour, the proposed formulation obtains the failure-time distribution as a result of cumulative damage dynamics that are discussed in the context of a geometric stochastic diffusion process. The model stresses, loads, velocities, misalignment, and environmental influences as a part of a multiplicative degradation kernel and directly introduces imperfect maintenance attenuation into the drift structure to provide a physically interpretable and analytically tractable model of conveyor belt wear-out behaviour.

The geometric diffusion was analytically transformed into logarithmic space to be able to derive a closed expression of the lognormal reliability in terms of first-passage principles. The resulting reliability function meets both theoretical condition of probabilistic boundedness, dimensional consistency, and stochastic well-posedness, and it can be used together with the classical exponential or Weibull-type behaviour under limiting conditions. The numerical findings were indicative of consistent wear-out behaviour, such as linear hazard increase and distinct transition of stable operation to accelerated degradation. The nearly equal value of the median failure time and the numerically integrated value of the MTTF showed moderate spreading and numerical stability of the formulation.

The sensitivity analysis revealed that the degradation growth parameter has a prevailing effect on the service life, which supports the need to operate the belt, to choose materials, and mitigate the environment to prolong the service life. The added stochastic uncertainty by diffusion terms and simulation-based confidence bands further facilitated the truthfulness of the reliability test by measuring variability in the long-term predictions of performance.

The combination of reliability modeling and preventive replacement optimization proved the practicality of the framework in terms of supporting maintenance decision making. The initial intervention age that was identified before the median failure time coincides with the accelerated hazard growth, thus balancing cost of lifecycle, operational risks, and availability performance. The associated steady-state availability of about 90% is an assurance that the suggested maintenance policy realizes stable operational ability coupled with evading the high-risk wear-out regime.

Altogether, the provided framework forms the consistent framework of degradation-reliability-maintenance-based approach specific to conveyor belt systems that are to ex-

perience accumulative abrasive wear and flawed maintenance. The resultant connection between physically interpretable damage evolution and probabilistic reliability theory and maintenance optimization, gives the study a stringent basis to predictive reliability evaluation and reliability-based maintenance planning in industrial bulk material handling settings. Future research can aim at estimating parameters based on field data and integration of condition monitoring in real time in order to improve model calibration and prediction.

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